

# A dissipative anomaly in the Hermite cascade of driven kinetic reduced MHD turbulence

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## Abstract

We report direct numerical evidence for a dissipative anomaly in the phase-space cascade of driven kinetic reduced magnetohydrodynamic (KRMHD) turbulence. Six simulations at  $128^3$  spatial and  $M = 128$  Hermite resolution, driven by Alfvénic forcing plus a low- $k_\perp$  Hermite source, were run for  $200 \tau_A$  each across  $\nu \in \{1, 3, 5, 10, 20, 50\}$  — a  $50\times$  range in collisionality. The time-averaged collisional dissipation rate  $\bar{\varepsilon}_\nu = 49.09 \pm 0.30$  is independent of  $\nu$  to better than 1%, while the Hermite spectrum  $\langle W(m) \rangle_t$  follows the Zocco–Schekochihin  $m^{-1/2}$  phase-mixing prediction in the inertial range  $m \in [4, 40]$  (measured slopes  $-0.472$  to  $-0.496$ ). At the dissipation scales,  $W(m)$  and  $\nu$  self-adjust so that the compensated quantity  $2\nu(m/M)^6 W(m)$  remains nearly invariant, with cumulative dissipation matching to  $< 2\%$  across the scan. The result confirms the phase-space analogue of the classical hydrodynamic dissipative anomaly and establishes a direct nonlinear benchmark for KRMHD solvers.

Reaching this regime required resolving a numerical instability in the default Lawson-RK4 Hermite integrator at large  $M \cdot k_z$ . The fix — an ARS(2,2,2) IMEX-RK222 scheme with implicit streaming and damping — is described in Appendix A and ships in GANDALF v0.5.0.

## 1 Introduction

The classical dissipative anomaly in hydrodynamic turbulence is the statement that the mean energy dissipation rate  $\varepsilon$  remains finite as the viscosity  $\nu \rightarrow 0$ , at fixed injection. This was conjectured by Onsager (1949) and is a central empirical fact of high-Reynolds-number turbulence (Duchon and Robert, 2000). The anomaly exists because the turbulent cascade delivers energy to progressively smaller scales at a rate that self-adjusts so the flux through any intermediate scale is  $\nu$ -independent; the dissipation scale itself moves with  $\nu$ , but the net dissipation does not.

The same logic ought to apply to any system that has a cascade and a dissipation mechanism acting at the far end of that cascade. In weakly collisional plasmas, KRMHD (Schekochihin et al., 2009) predicts a phase-space cascade in which the perturbed distribution function, expanded in Hermite moments  $g_m$  of the parallel velocity, transfers energy from low  $m$  to high  $m$  under parallel streaming. Collisions act as a dissipation mechanism concentrated at high  $m$ . The obvious question is whether  $\varepsilon_\nu$ , the collisional dissipation rate computed from the Hermite spectrum, approaches a  $\nu$ -independent plateau as  $\nu \rightarrow 0$ .

Linear phase mixing, in which  $g_m$  cascades deterministically under the streaming operator alone, was analysed by Zocco and Schekochihin (2011) and gives a Hermite spectrum  $W(m) \propto m^{-1/2}$ . The nonlinear problem, in which turbulent Alfvénic fluctuations at  $z^\pm$  nonlinearly couple to  $g_m$  through the Poisson bracket  $\{\Phi, g_m\}$ , is much harder and has been a standing target since Kanekar (2015). Prior numerical attempts at moderate resolution have not cleanly separated an inertial range from the forcing and dissipation scales, nor directly demonstrated a  $\nu$ -independent  $\varepsilon_\nu$ .

In this paper we present such a demonstration. At  $128^3$  spatial resolution with  $M = 128$  Hermite moments and hyper-collisional damping  $\propto (m/M)^6$ , we drive Alfvénic turbulence to a steady state and then add a low- $k_\perp$  forcing on the  $g_0$  moment. After a  $\sim 25 \tau_A$  fill-in transient the Hermite sector settles into a statistically stationary cascade. Scanning  $\nu$  over a factor of 50 produces  $\bar{\varepsilon}_\nu$  values that agree to  $< 1\%$ , with an inertial-range  $W(m)$  slope within  $\sim 2\%$  of  $m^{-1/2}$  and a visible compensation between  $W$  and  $\nu$  at the damped scales. We interpret this as direct numerical evidence for the dissipative anomaly in a kinetic plasma turbulence problem.

An ancillary but practically important contribution is the diagnosis and fix of a scheme-level numerical instability that had previously prevented reaching this regime. The Lawson-RK4 integrator used by default in GANDALF composes an exact integrating factor for linear streaming with explicit RK stages on the nonlinear advection, which leaks a hidden CFL condition at large  $M \cdot k_z$  into the stability envelope and triggers a simultaneous all- $m$  exponential blowup after  $\mathcal{O}(100) \tau_A$ . An IMEX-RK222 integrator (Ascher et al., 1997) that treats linear streaming and hyper-collisional damping implicitly resolves the instability; it ships in GANDALF v0.5.0 (Kanekar, 2026). A brief account is given in Appendix A.

## 2 Model and numerics

We solve the KRMHD equations (Schekochihin et al., 2009) on a triply periodic domain of size  $L_x = L_y = L_z = 1$  with  $v_A = 1$ ,  $\beta_i = 1$ , and ion-to-electron temperature ratio  $\tau = 1$ . The state consists of the Elsasser fields  $z^\pm$ , the parallel magnetic-field perturbation  $B_\parallel$ , and the first  $M + 1$  Hermite moments  $g_m$  of the ion perturbed distribution function, with  $M = 128$ . The coupling coefficient between the fluid and Hermite sectors is  $\Lambda = \sqrt{5}$ , obtained from the KRMHD dispersion relation at our parameters; it is essential to override the default  $\Lambda = 1$  carried in Alfvénic-only checkpoints, which silently zeros the  $g_0 \rightarrow g_1$  coupling through the  $(1 - 1/\Lambda)$  prefactor.

Spatial resolution is  $128^3$  with  $2/3$  dealiasing. Forcing is applied in two channels. The Alfvénic drive is a balanced Elsasser white-noise forcing restricted to  $n_\perp \in [1, 2]$  and  $|n_z| = 1$  (excluding  $n_z = 0$ ), with amplitude  $f = 0.02$ , following the low- $k_z$  restriction necessary to avoid RMHD-incompatible high- $k_z$  injection. The Hermite drive acts on the  $g_0$  moment only, with the same perpendicular and parallel wavenumber mask as the Alfvénic drive and amplitude  $f_H = 3.5 \times 10^{-3}$ . Resistive damping is  $\eta = 100$  with hyper-order  $r = 2$ ; Hermite damping is hyper-collisional,  $\nu_m = \nu(m/M)^6$ , active for  $m \geq 2$ .

Collisional dissipation is computed from the Hermite spectrum as

$$\varepsilon_\nu(t) = 2\nu \sum_{m=2}^M \left(\frac{m}{M}\right)^6 W(m, t), \quad W(m, t) = \sum_{\mathbf{k}} |g_m(\mathbf{k}, t)|^2. \quad (1)$$

Time-averaged quantities  $\bar{\varepsilon}_\nu$ ,  $\langle W(m) \rangle_t$  are taken over  $t - t_0 > 30 \tau_A$  to exclude the Hermite fill-in transient, where  $t_0$  is the time at which  $g_0$  forcing is activated.

Time integration uses the IMEX-RK222 scheme of Ascher et al. (1997) as implemented in GANDALF v0.5.0 (Kanekar, 2026). Linear streaming and hyper-collisional damping are solved implicitly via a per- $k_z$  batched LU factorisation on the Hermite axis; the Poisson-bracket nonlinearities remain explicit. The Elsasser subsystem is advanced by an integrating-factor RK2 midpoint rule that is bit-identical to the prior Lawson-RK4 path. The timestep  $dt = 2.34 \times 10^{-3}$  is set by the perpendicular advection CFL with safety 0.3. We verify the solver by running a linear Hermite benchmark with no Alfvénic forcing: at  $M = 128$ ,  $\nu = 1$ , the predicted  $m^{-1/2}$  phase-mixing spectrum develops over  $500 \tau_A$  with  $\bar{\varepsilon}_\nu \approx 0.23$ , consistent with Zocco and Schekochihin (2011).

Table 1: Time-averaged dissipation, total Hermite energy, and inertial-range slope for the six IMEX runs. Time-series mean and spectrum-window mean are taken over  $t - t_0 > 30 \tau_A$  and  $t \in [2100, 2200] \tau_A$  respectively.

| $\nu$ | $\bar{\varepsilon}_\nu$ (ts) | $\bar{\varepsilon}_\nu$ (spec) | $\langle \sum_m W(m) \rangle_t$ | slope $m \in [4, 40]$ |
|-------|------------------------------|--------------------------------|---------------------------------|-----------------------|
| 1     | $48.42 \pm 11.41$            | 48.56                          | 359                             | -0.472                |
| 3     | $49.21 \pm 10.65$            | 48.15                          | 162                             | -0.484                |
| 5     | $49.26 \pm 10.61$            | 47.99                          | 137                             | -0.489                |
| 10    | $49.23 \pm 10.81$            | 47.99                          | 126                             | -0.493                |
| 20    | $49.21 \pm 10.99$            | 48.26                          | 121                             | -0.494                |
| 50    | $49.21 \pm 11.21$            | 48.79                          | 114                             | -0.496                |

All nonlinear runs resume from a common Alfvénic steady-state checkpoint at  $t_0 = 2000 \tau_A$ , obtained by driving a pure  $M = 0$  RMHD run to an 8%-variation steady state under the same forcing and damping. Each run extends for  $200 \tau_A$  of new simulated time, with  $g_0$  forcing activated at  $t_0$ . Diagnostics are saved every 100 steps; Hermite spectra are saved every 500 steps during the averaging window  $t \in [2100, 2200] \tau_A$ .

## 3 Results

### 3.1 The $\varepsilon_\nu(\nu)$ plateau

Figure 1 shows the time-averaged collisional dissipation  $\bar{\varepsilon}_\nu$  as a function of  $\nu$  across six runs. The six values (Table 1) cluster at  $\bar{\varepsilon}_\nu = 49.09 \pm 0.30$ ; the full range is 48.42 to 49.26, spanning  $< 2\%$ . Across a factor of 50 in  $\nu$  there is no systematic trend: the plateau is flat. The spectrum-window mean (red diamonds in Figure 1, left panel) agrees with the full time-series mean to  $\lesssim 1\%$ , indicating the averaging is well converged.

The right panel of Figure 1 shows the mechanism. As  $\nu$  decreases, the Hermite sector accumulates more total energy —  $\langle \sum_m W(m) \rangle_t$  rises from 114 at  $\nu = 50$  to 359 at  $\nu = 1$ . The relationship has a visible kink around  $\nu \approx 3$ ; above that value the energy depends weakly on  $\nu$ , below it the dependence steepens. What does not change is the integrated dissipation. This is the compensating behaviour expected of an anomaly: the spectrum adjusts at the damped scales so that the product  $\nu \cdot W$  gives the same dissipation regardless of the prefactor.

### 3.2 The Hermite spectrum and compensated integrand

Figure 2 shows the time-averaged Hermite spectrum  $\langle W(m) \rangle_t$  on the left and the per- $m$  dissipation integrand  $2\nu(m/M)^6 W(m)$  on the right, for all six runs. The inertial range  $m \in [4, 40]$  is essentially degenerate across  $\nu$  on the left panel, and a log-log fit in that range returns slopes  $-0.472$  (at  $\nu = 1$ ) to  $-0.496$  (at  $\nu = 50$ ), with the outermost three values at  $\nu = 10, 20, 50$  within 1% of the Zocco–Schekochihin  $m^{-1/2}$  prediction. The dashed black reference line in the figure is  $m^{-1/2}$  anchored at the  $\nu = 3$  curve.

At  $m \gtrsim 60$  the curves split in the expected order: lower  $\nu$  carries more energy in the damped tail, because weaker collisional damping lets energy pile up before the  $(m/M)^6$  factor catches it. The right panel of Figure 2 shows the corresponding dissipation integrand. At  $\nu = 50$  the integrand peaks at  $m \approx 80$ ; at  $\nu = 1$  it peaks near  $m = M$ . But the areas under all six curves are equal to  $\sim 2\%$ : the cumulative values are  $\varepsilon_\nu \in \{48.6, 48.1, 48.0, 48.0, 48.3, 48.8\}$  (ordered by increasing

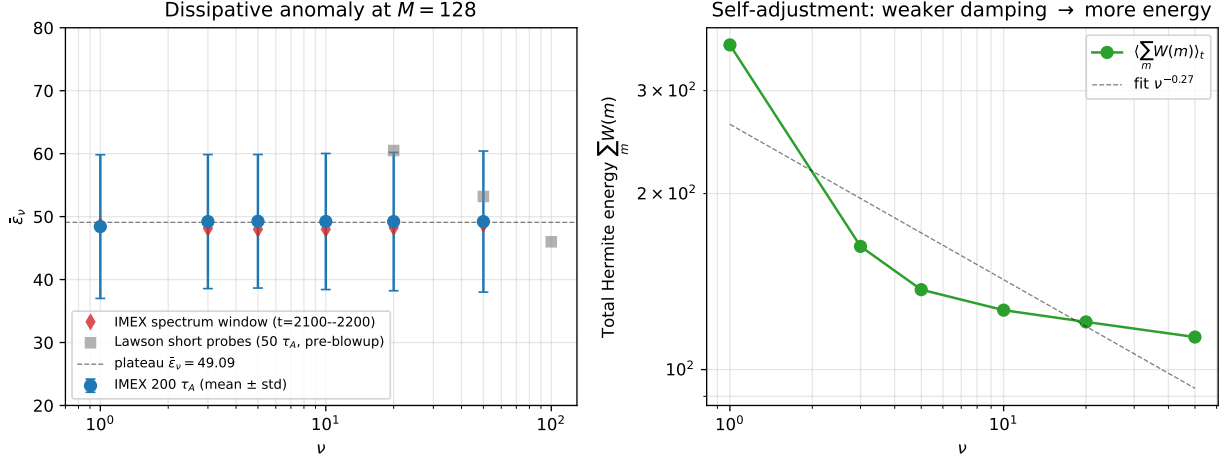


Figure 1: **Dissipative anomaly.** Left: time-averaged collisional dissipation  $\bar{\varepsilon}_\nu$  as a function of  $\nu$ , from six IMEX runs at  $M = 128, 128^3, 200 \tau_A$ . Blue circles are time-series means with  $\pm 1\sigma$  error bars; red diamonds are spectrum-window means. The horizontal dashed line at  $\bar{\varepsilon}_\nu = 49.09$  is the six-point mean. Gray squares are legacy Lawson-RK4 short-probe points ( $50 \tau_A$  before numerical blowup set in; see Appendix A) and are shown for reference only. Right: total time-averaged Hermite energy  $\langle \sum_m W(m) \rangle_t$  versus  $\nu$ , showing the self-adjustment of the spectrum: as  $\nu$  decreases, more total Hermite energy is carried (by  $\sim 3\times$  from  $\nu = 50$  to  $\nu = 1$ ), while  $\bar{\varepsilon}_\nu$  stays fixed.

$\nu$ ). This is the dissipative anomaly at the spectral level — the dissipation range moves with  $\nu$ , the spectrum within it adjusts, and the integral that matters for the energy budget remains invariant.

## 4 Discussion

## Acknowledgements

## A A numerical instability in Lawson-RK4 and the IMEX-RK222 fix

## References

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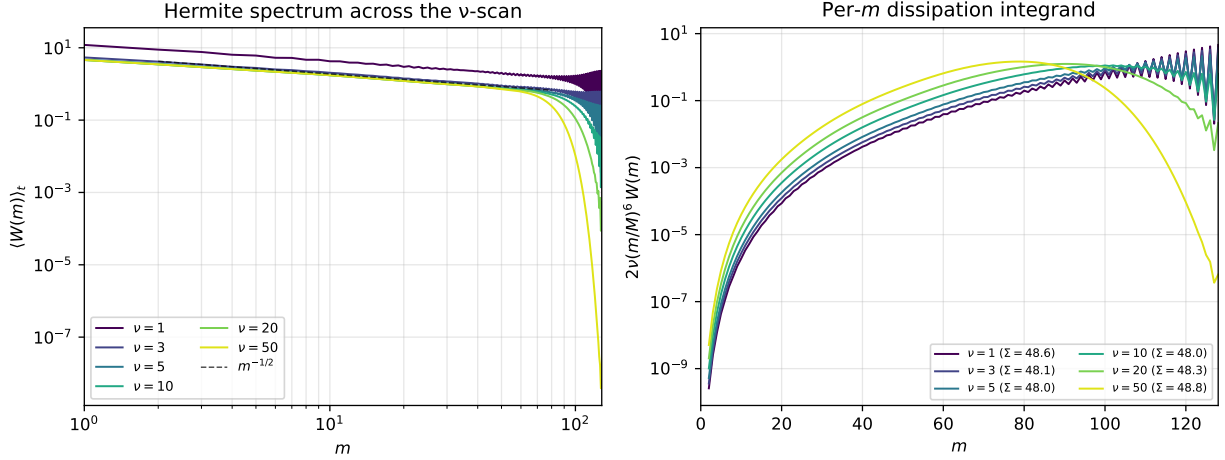


Figure 2: **Hermite spectrum and compensated dissipation integrand.** Left: time-averaged  $\langle W(m) \rangle_t$  across the six runs, log-log, over the averaging window  $t - t_0 > 100 \tau_A$ . Colours run from  $\nu = 1$  (purple) to  $\nu = 50$  (yellow). The black dashed line is the Zocco–Schekochihin prediction  $W(m) \propto m^{-1/2}$ . Inertial-range slopes  $m \in [4, 40]$  are within 5% of this line across the scan. Right: per- $m$  dissipation integrand  $2\nu(m/M)^6 W(m)$ . The peak shifts from near  $m = M$  at low  $\nu$  to  $m \approx 80$  at  $\nu = 50$ , but the area under each curve (cumulative  $\varepsilon_\nu$ , legend values) matches to  $< 2\%$ .

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